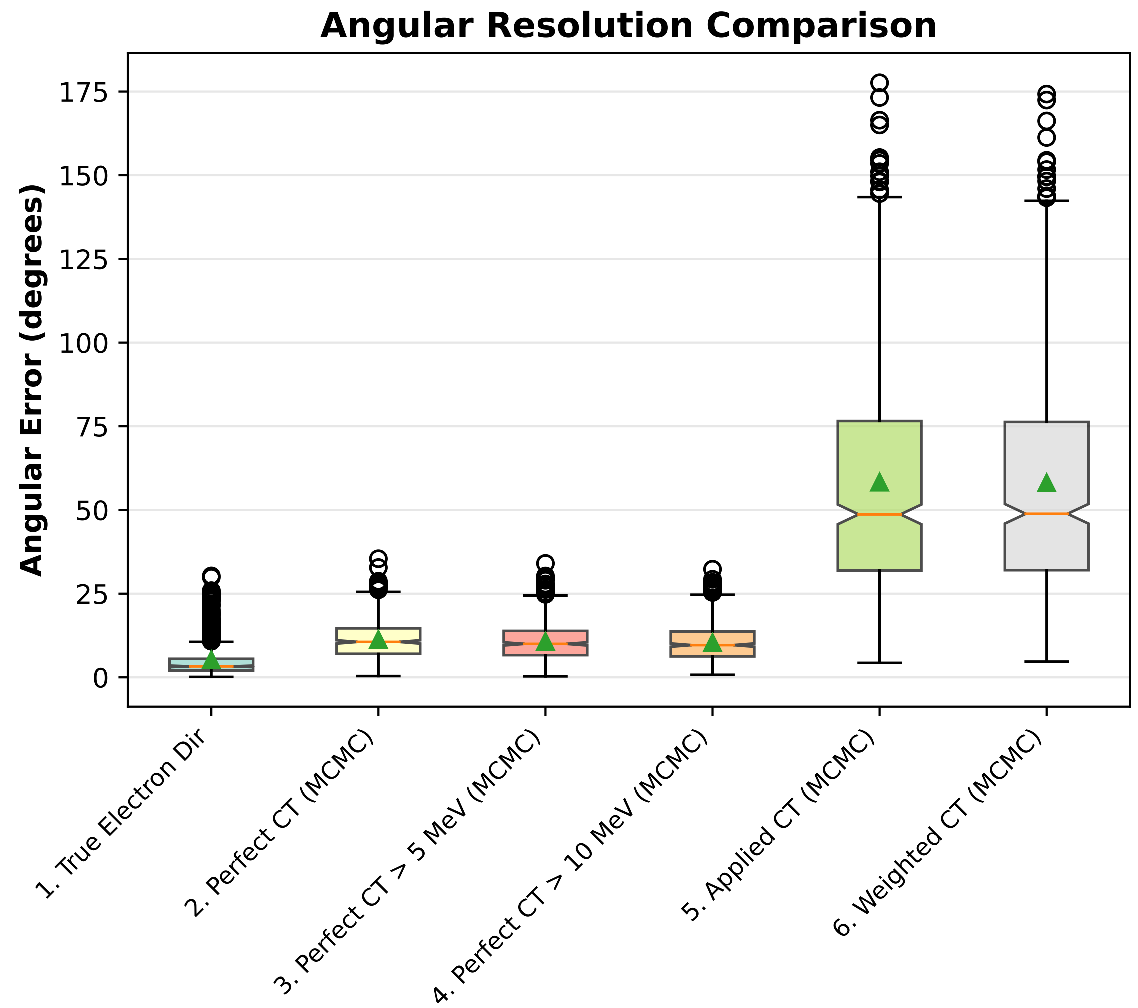
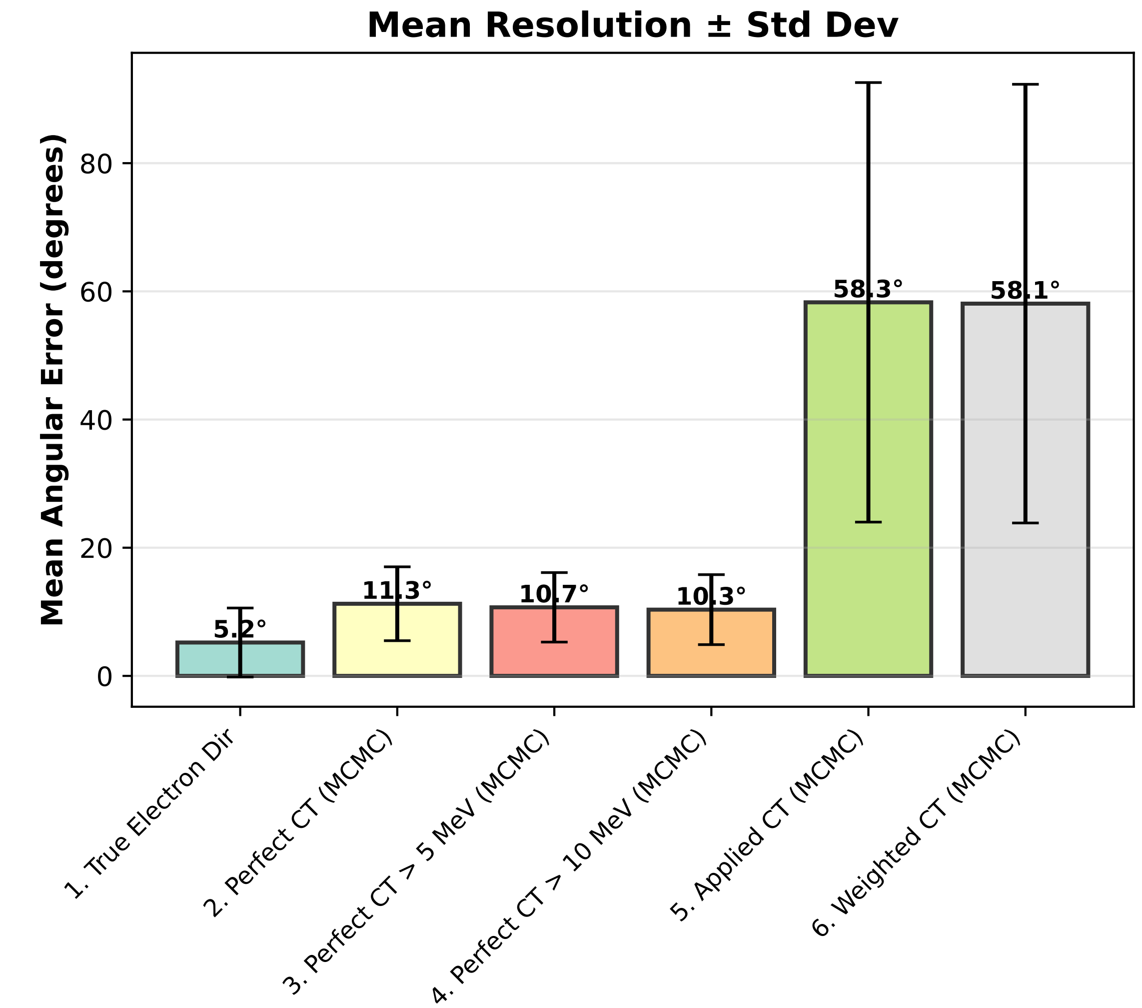
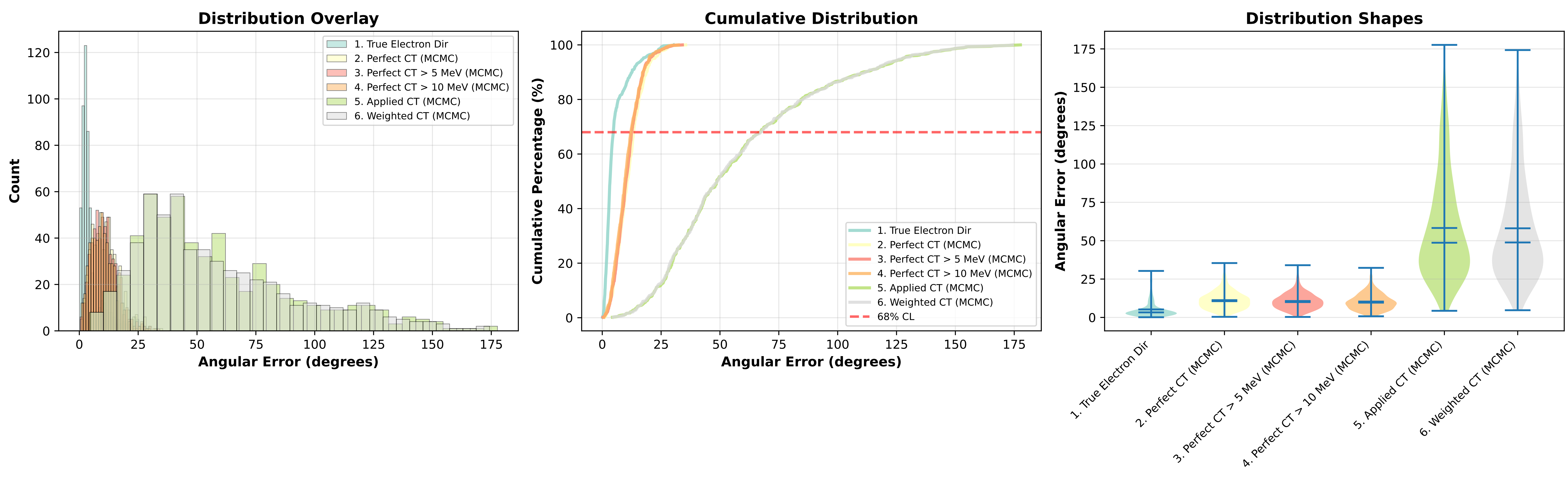
Emcee 8-Scenario Aggregate Analysis
Avg ES clusters: 1.: 271 ES | 2.: 271 ES | 3.: 241 ES | 4.: 147 ES
5.: 3242 ES | 6.: 3242 ES

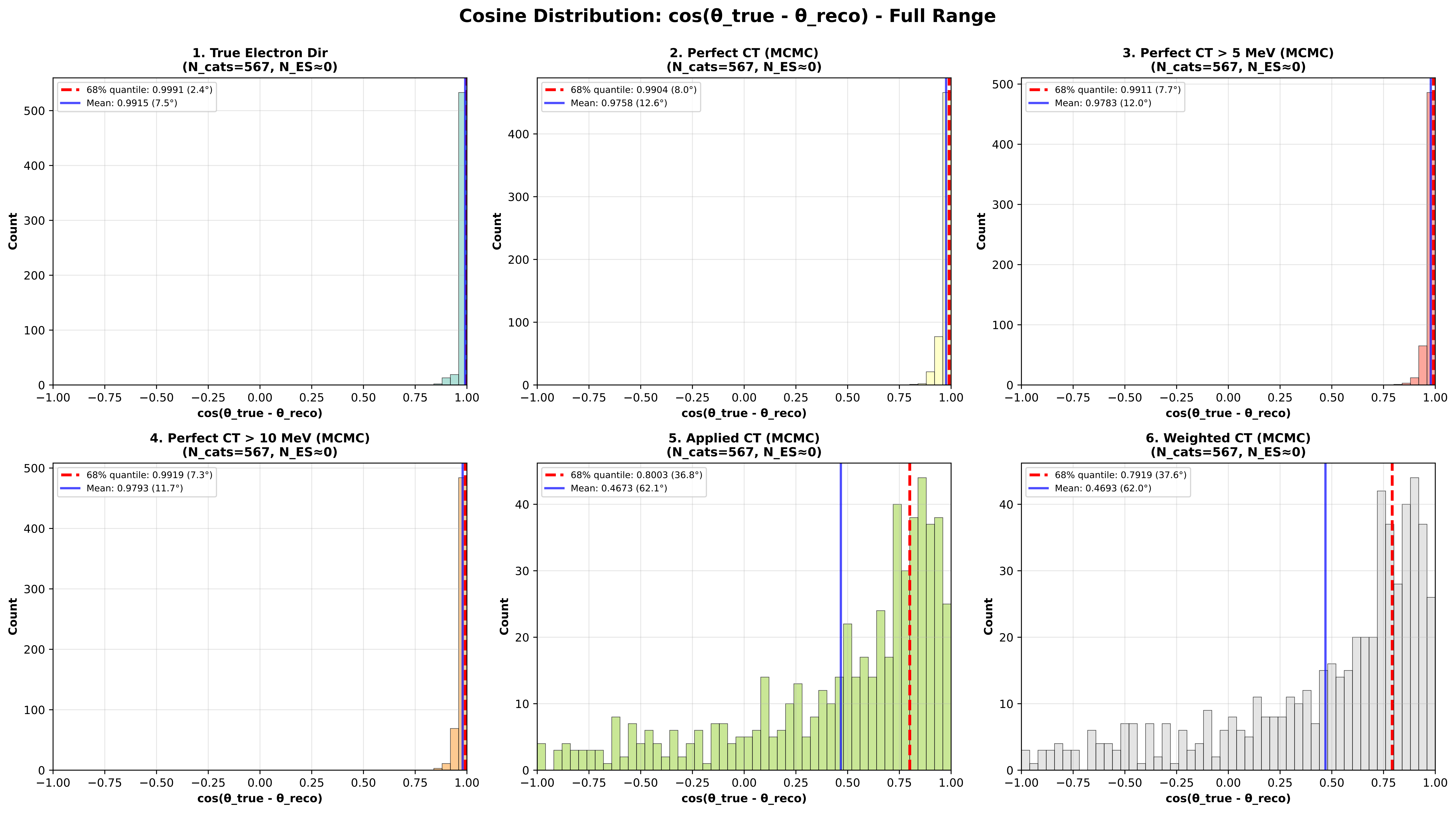


Summary Statistics

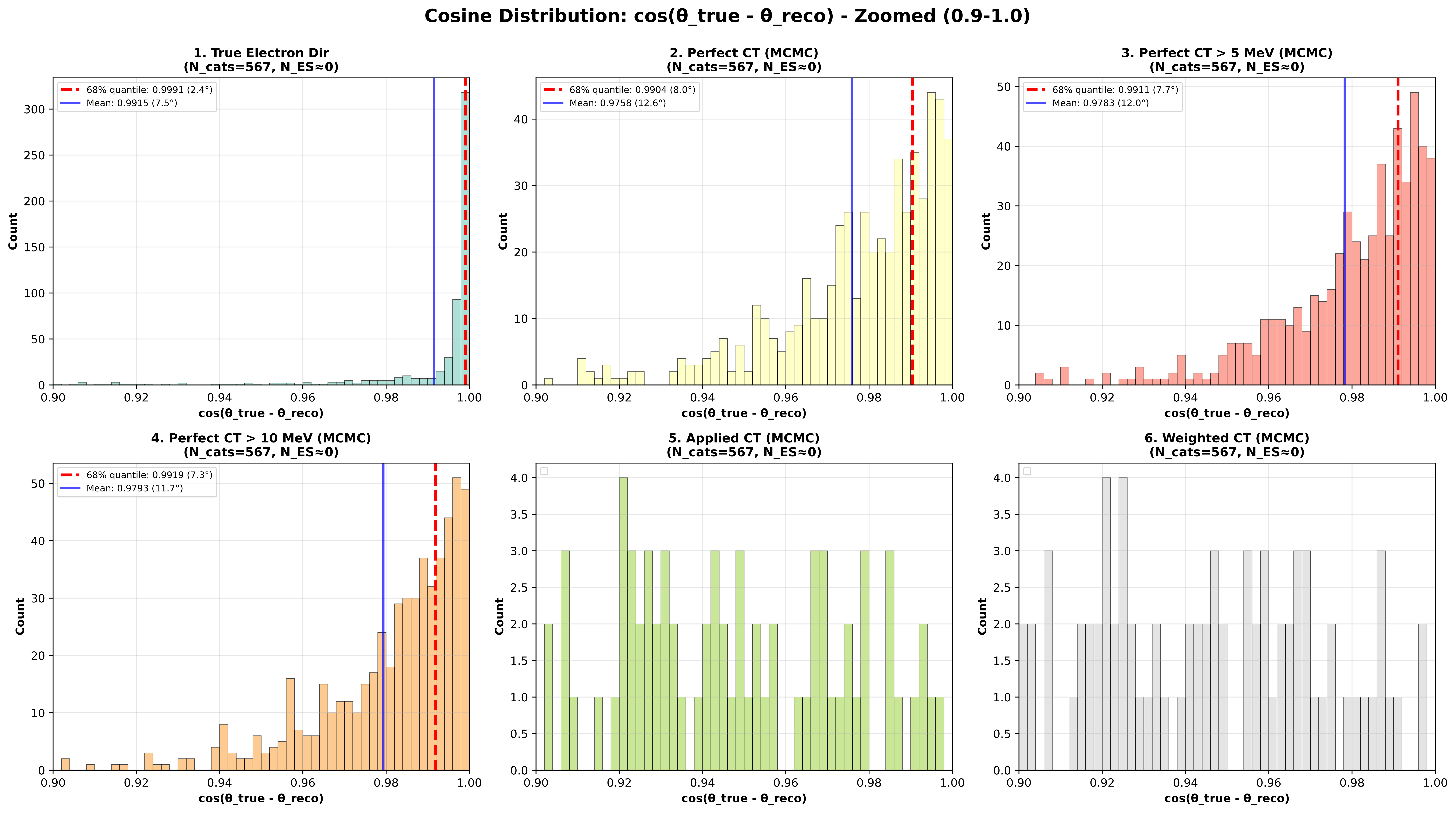
Scenario	N	68% Q	Median	Mean
1. True Electror	567	4.72°	3.28°	5.21°
2. Perfect CT (M	567	13.25°	10.58°	11.26°
3. Perfect CT >	567	12.54°	10.00°	10.70°
4. Perfect CT >	567	12.17°	9.64°	10.34°
5. Applied CT (	567	67.16°	48.68°	58.29°
6. Weighted CT	567	67.21°	48.86°	58.08°







Cosine Distribution: cos(θ_true - θ_reco) - Full Range
1. True Electron Dir
(N_cats=567, N_ES≈0)
68% quantile: 0.9991 (2.4°)
Mean: 0.9915 (7.5°)
2. Perfect CT (MCMC)
(N_cats=567, N_ES≈0)
68% quantile: 0.9904 (8.0°)
Mean: 0.9758 (12.6°)
3. Perfect CT > 5 MeV (MCMC)
(N_cats=567, N_ES≈0)
68% quantile: 0.9911 (7.7°)
Mean: 0.9783 (12.0°)
4. Perfect CT > 10 MeV (MCMC)
(N_cats=567, N_ES≈0)
68% quantile: 0.9919 (7.3°)
Mean: 0.9793 (11.7°)
5. Applied CT (MCMC)
(N_cats=567, N_ES≈0)
68% quantile: 0.8003 (36.8°)
Mean: 0.4673 (62.1°)
6. Weighted CT (MCMC)
(N_cats=567, N_ES≈0)
68% quantile: 0.7919 (37.6°)
Mean: 0.4693 (62.0°)
Count
cos(θ_true - θ_reco)

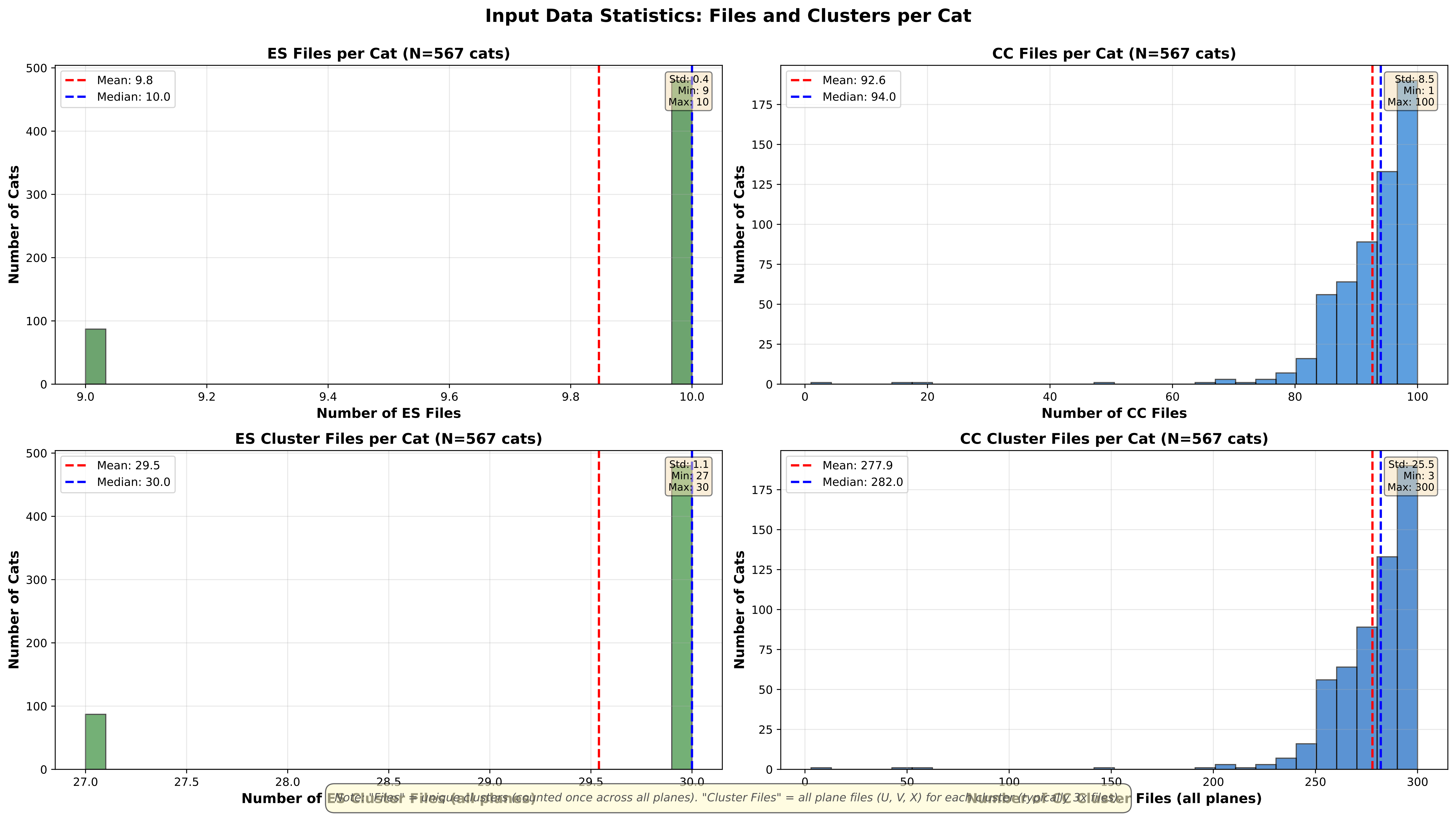

Cosine Distribution: cos(θ_true - θ_reco) - Zoomed (0.9-1.0)
1. True Electron Dir (N_cats=567, N_ES≈0)
68% quantile: 0.9991 (2.4°)
Mean: 0.9915 (7.5°)
2. Perfect CT (MCMC) (N_cats=567, N_ES≈0)
68% quantile: 0.9904 (8.0°)
Mean: 0.9758 (12.6°)
3. Perfect CT > 5 MeV (MCMC) (N_cats=567, N_ES≈0)
68% quantile: 0.9911 (7.7°)
Mean: 0.9783 (12.0°)
4. Perfect CT > 10 MeV (MCMC) (N_cats=567, N_ES≈0)
68% quantile: 0.9919 (7.3°)
Mean: 0.9793 (11.7°)
5. Applied CT (MCMC) (N_cats=567, N_ES≈0)
6. Weighted CT (MCMC) (N_cats=567, N_ES≈0)
Count
cos(θ_true - θ_reco)

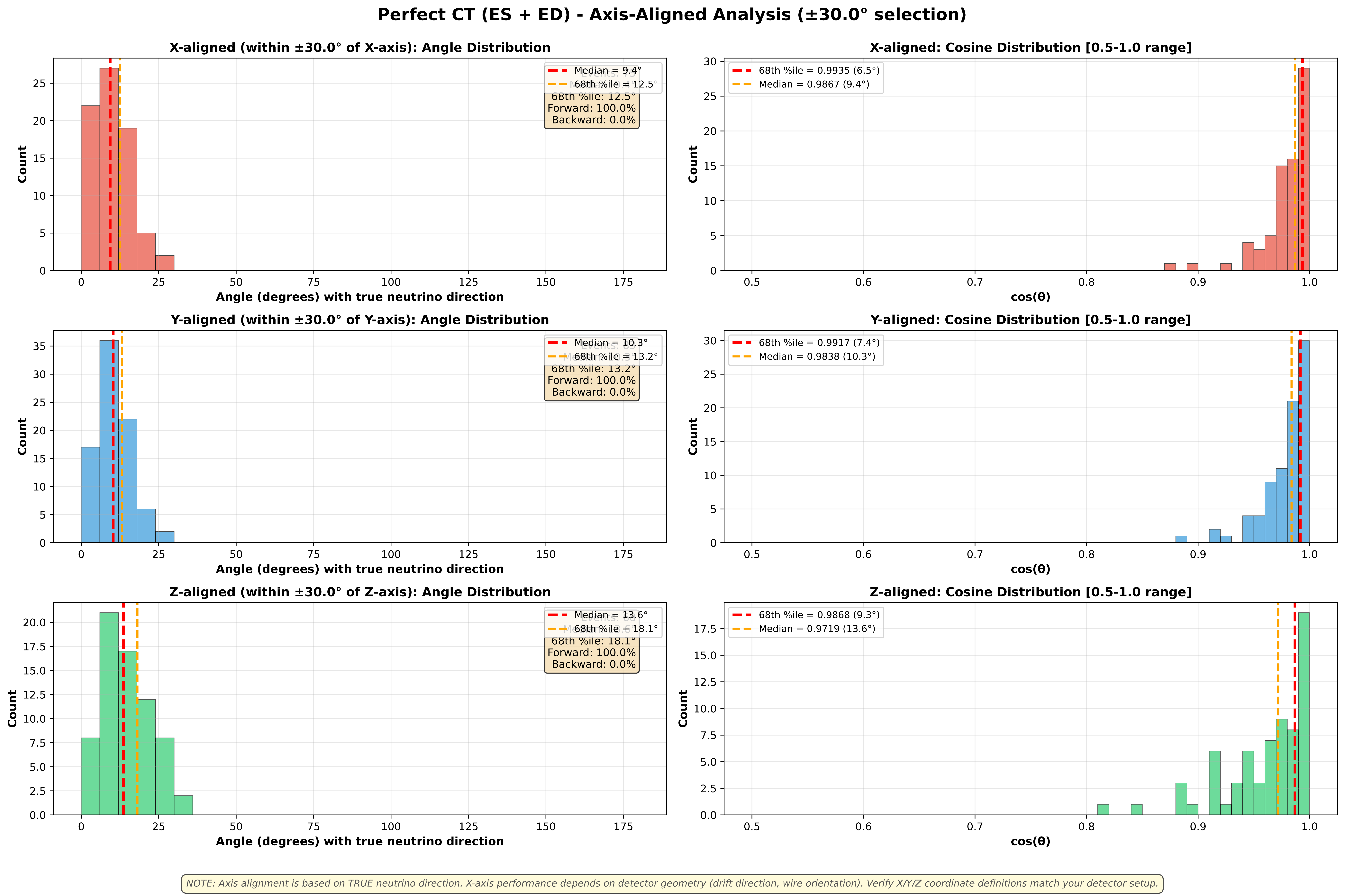
Cluster Usage Across Scenarios

Scenario	N Cats	Input Total	Input ES Main	Value "Clusters Used"	Notes
1. True Electron Dir	567	102	10	271	MCMC samples (weighted sum)
2. Perfect CT (MCMC)	567	102	10	271	MCMC samples (weighted sum)
3. Perfect CT > 5 MeV (MCMC)	567	102	10	241	MCMC samples (weighted sum)
4. Perfect CT > 10 MeV (MCMC)	567	102	10	147	143.6% retention
5. Applied CT (MCMC)	567	102	10	3242	MCMC samples (weighted sum)
6. Weighted CT (MCMC)	567	102	10	3242	MCMC samples (weighted sum)

Note: "Clusters Used" shows different meanings per scenario:
• Best Case, Perfect CT: Number of ES main clusters actually used (after E>3 MeV cut)
• Full Pipeline, Weighted CT: Weighted sum of cluster contributions in MCMC (includes CT probabilities)
• Energy cut scenarios: Number of clusters passing energy threshold

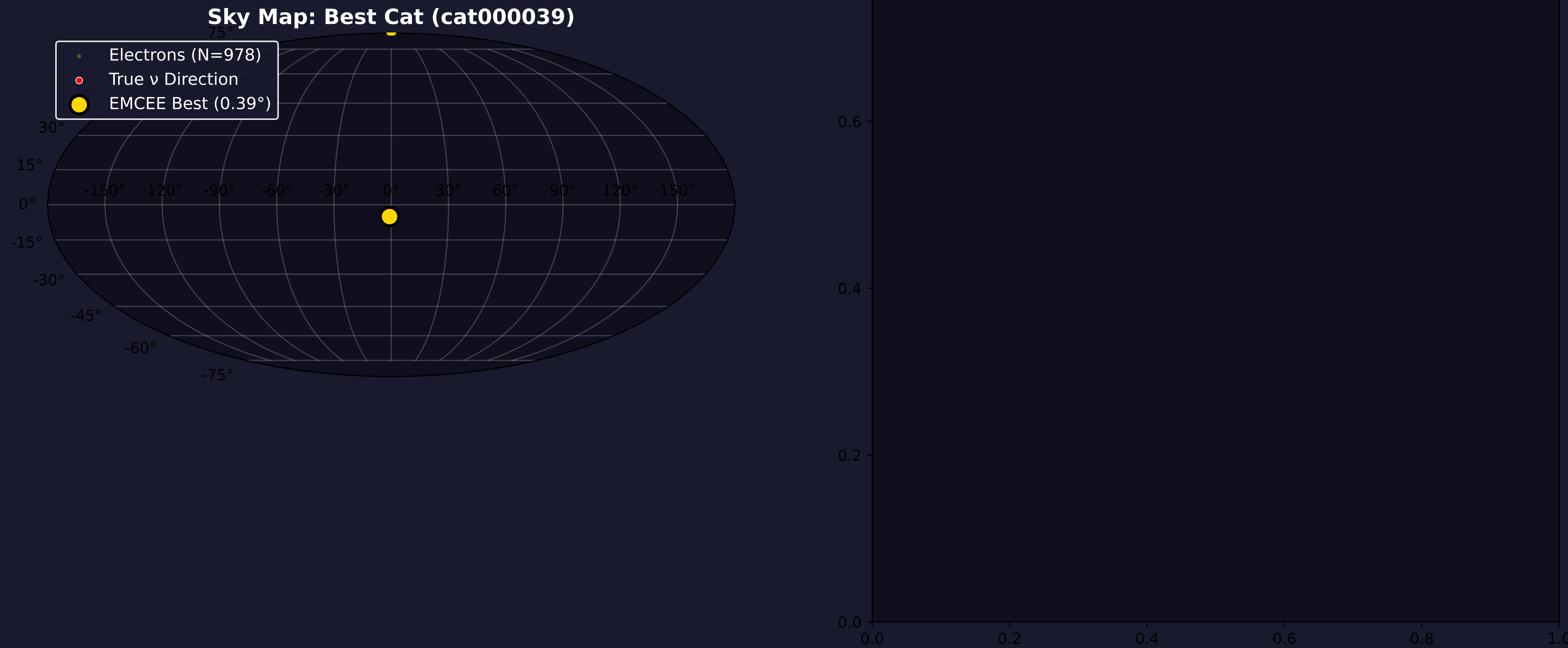
The key insight: Best Case and Perfect CT use ~85-90% of ES main tracks with E>3 MeV.
Full Pipeline shows much higher values due to weighted MCMC sampling, not actual cluster count.


Input Data Statistics: Files and Clusters per Cat
ES Files per Cat (N=567 cats)
Mean: 9.8
Median: 10.0
Std: 0.4
Min: 9
Max: 10
Number of ES Files
Number of Cats
CC Files per Cat (N=567 cats)
Mean: 92.6
Median: 94.0
Std: 8.5
Min: 1
Max: 100
Number of CC Files
Number of Cats
ES Cluster Files per Cat (N=567 cats)
Mean: 29.5
Median: 30.0
Std: 1.1
Min: 27
Max: 30
Number of ES Cluster Files (all planes)
Number of Cats
CC Cluster Files per Cat (N=567 cats)
Mean: 277.9
Median: 282.0
Std: 25.5
Min: 3
Max: 300
Number of CC Cluster Files (all planes)
Number of Cats
Note: "Files" = Unique clusters (counted once across all planes). "Cluster Files" = all plane files (U, V, X) for each cluster (typically 3 files).

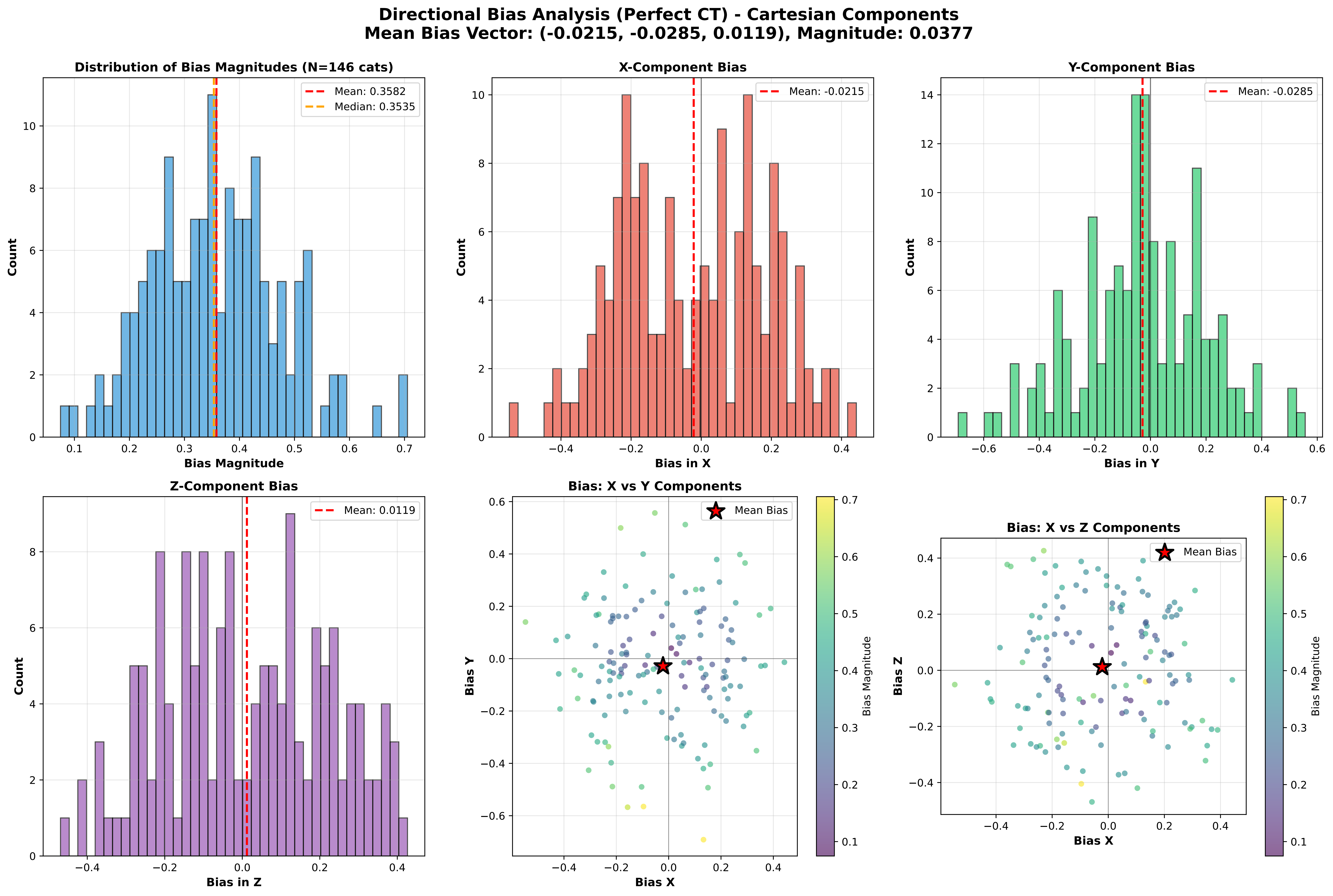


Perfect CT (ES + ED) - Axis-Aligned Analysis (±30.0° selection)
X-aligned (within ±30.0° of X-axis): Angle Distribution
Median = 9.4°
68th %ile = 12.5°
68th %ile: 12.5°
Forward: 100.0%
Backward: 0.0%
Angle (degrees) with true neutrino direction
Count
X-aligned: Cosine Distribution [0.5-1.0 range]
68th %ile = 0.9935 (6.5°)
Median = 0.9867 (9.4°)
cos(θ)
Y-aligned (within ±30.0° of Y-axis): Angle Distribution
Median = 10.3°
68th %ile = 13.2°
68th %ile: 13.2°
Forward: 100.0%
Backward: 0.0%
Y-aligned: Cosine Distribution [0.5-1.0 range]
68th %ile = 0.9917 (7.4°)
Median = 0.9838 (10.3°)
Z-aligned (within ±30.0° of Z-axis): Angle Distribution
Median = 13.6°
68th %ile = 18.1°
68th %ile: 18.1°
Forward: 100.0%
Backward: 0.0%
Z-aligned: Cosine Distribution [0.5-1.0 range]
68th %ile = 0.9868 (9.3°)
Median = 0.9719 (13.6°)
NOTE: Axis alignment is based on TRUE neutrino direction. X-axis performance depends on detector geometry (drift direction, wire orientation). Verify X/Y/Z coordinate definitions match your detector setup.

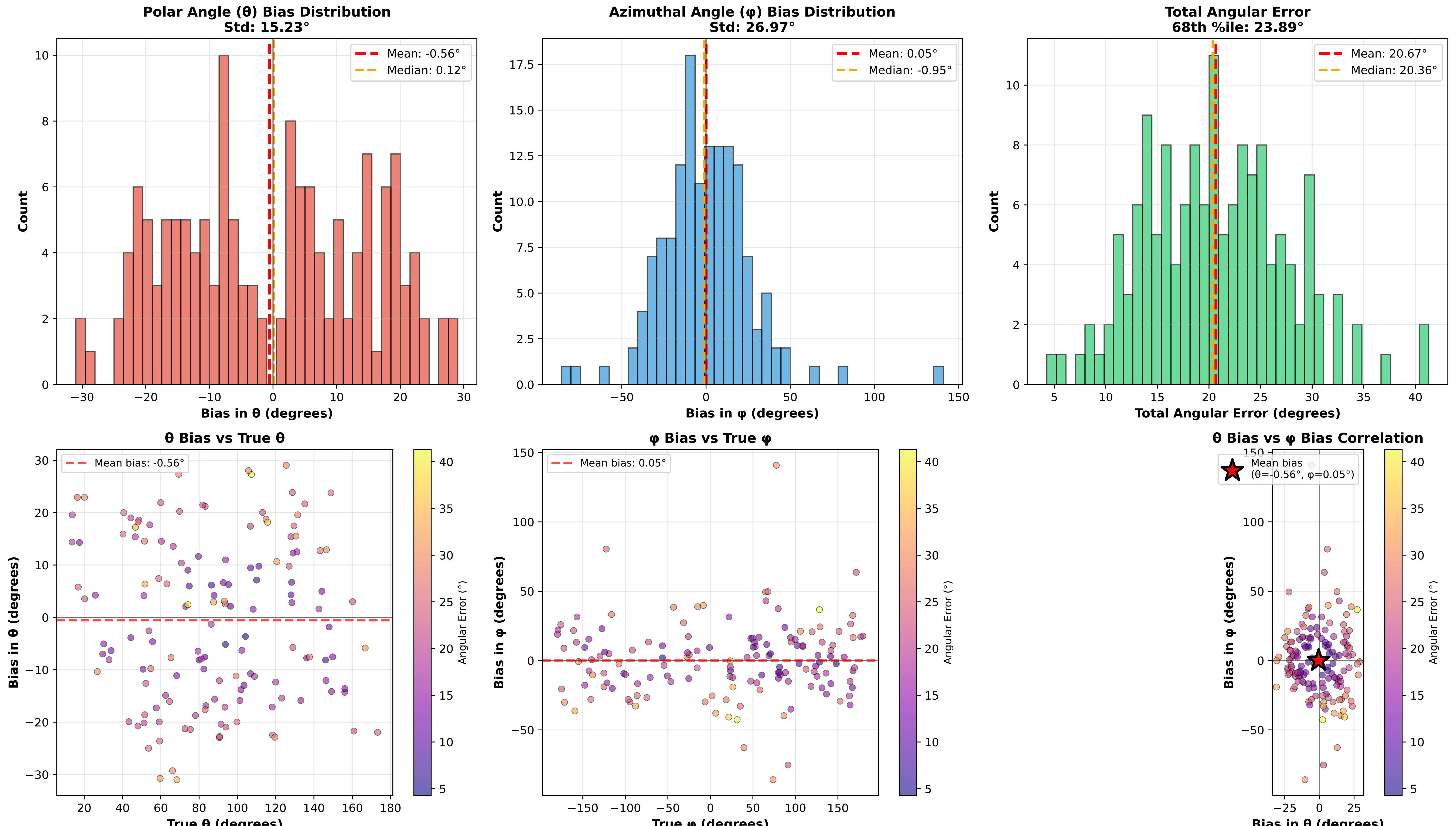
Neutrino Pointing Reconstruction: cat000039 - Perfect CT Scenario
Yellow: Electrons | Red: True Neutrino Direction



Best cat from EMCEE analysis: cat000039 | Angular Error: 0.39° | 68% Credible Interval (ω_{68}): 0.33° | N electrons: 200


Directional Bias Analysis (Perfect CT) - Cartesian Components
Mean Bias Vector: (-0.0215, -0.0285, 0.0119), Magnitude: 0.0377
Distribution of Bias Magnitudes (N=146 cats)
Mean: 0.3582
Median: 0.3535
Bias Magnitude
Count
X-Component Bias
Mean: -0.0215
Bias in X
Y-Component Bias
Mean: -0.0285
Bias in Y
Z-Component Bias
Mean: 0.0119
Bias in Z
Bias: X vs Y Components
Mean Bias
Bias X
Bias Y
Bias Magnitude
Bias: X vs Z Components
Mean Bias
Bias X
Bias Z
Bias Magnitude

Angular Bias Analysis in Spherical Coordinates (Perfect CT)
θ = polar angle from z-axis [0°, 180°], φ = azimuthal angle from x-axis [-180°, 180°]



Angular Bias Statistics (N=146 cats):
θ bias: mean=-0.56°, median=0.12°, std=15.23°
φ bias: mean=0.05°, median=-0.95°, std=26.97°
Total angular error: mean=20.67°, 68th %ile=23.89°